

Total no. of Pages:

Roll no.

B. Tech: ECE (VLSI Design and technology)
MID-SEMESTER EXAMINATION, September-2025

Semester: III

Course Title: Microelectronics
Course Code: VTCECC304

Duration: 1:30 Hours

Max. Marks: 20

Note: - Attempt all questions in the given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	In a BJT amplifier, the DC bias collector current is 1mA , the β of the transistor is 100 and the Early voltage is $V_A=100\text{ Volts}$, determine the values of the small signal parameters r_π , β_{ac} , r_o and r_e where the symbols have their usual meaning.	2	CO1
1b	For the MOSFET circuit of Fig. 1, deduce the expressions for the transconductance $G_m = \frac{i_o}{v_{in}}$ and the small signal output resistance r_{out} in terms of the parameters: β_m and r_o of the MOSFET and the source degeneration R_s .	2	CO2
2a	For the common emitter amplifier of Fig. 2, assume that all the three transistors are matched and have same $\beta=100$, and Early voltage $=100\text{ Volts}$. Determine the DC bias collector current of Q_1 and the small signal voltage gain of the amplifier.	2	CO4
2b	For the BJT circuit of Fig. 3, find the value of the small signal input resistance R_{in} , given that $I_{bias}=1\text{mA}$, $\beta=100$, $V_A=100\text{V}$ and $R_E=5\text{k}\Omega$.	2	CO1
3a	For the MOS cascode amplifier of fig. 4, show that the output resistance rout of the amplifier is given by: $r_{out} = r_{o2} (1 + \beta_{out} r_{o1})$, state the approximations/assumptions made, if any	2	CO2
3b	By appropriate analysis, determine the value of the small signal output impedance of the cascode current mirror of Fig. 5, assuming that all the transistors are operating in saturation, have same $K = \frac{1}{2} \mu_n C_{ox} (W/L) = 160 \mu\text{A/V}^2$ and carry dc drain currents of $20 \mu\text{A}$ and have Early voltage $V_A=10\text{V}$	2	CO4
4a	Determine an expression of the common-mode gain for the NMOS differential amplifier of Fig. 6 in terms of relevant parameters. Assume that M_1, M_2 are matched, and M_3, M_4 are matched and a single-ended output V_{o1} is of interest.	2	CO2
4b	By appropriate analysis, explain how the circuit of Fig. 7 can function as a differential to single-ended converter.	2	CO2
5a	Explain Miller's theorem and its dual	2	CO1
5b	Explain the parameters of the high-frequency model of a MOSFET. Draw the high frequency equivalent circuit of the CS amplifier of Fig. 8 and explain, the use of Miller's theorem in determining the dominant pole and the 3-dB bandwidth of this amplifier.	2	CO2

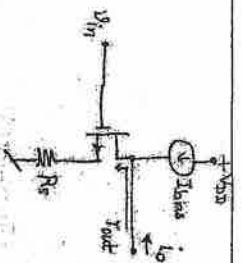


Fig. 1

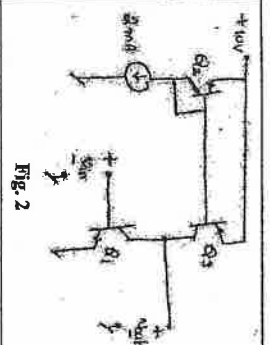


Fig. 2

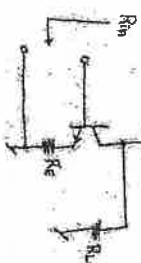


Fig. 3

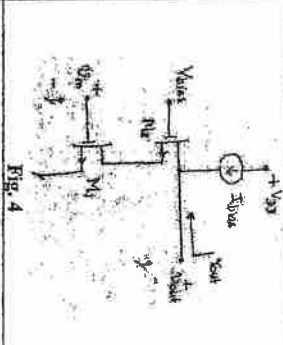


Fig. 4

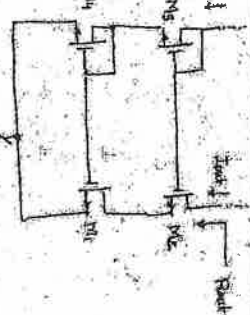


Fig. 5

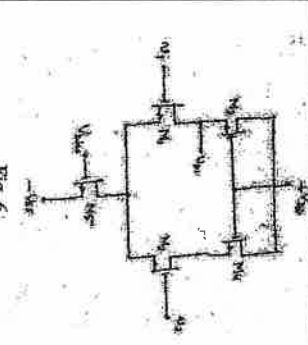


Fig. 6

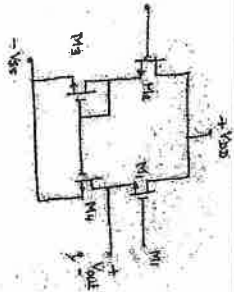


Fig. 7

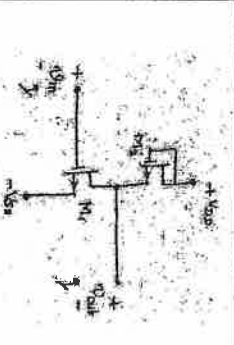


Fig. 8